

- Q1.** Which of the following relations is a function?
- (A) A relation with two inputs having the same output
(B) A relation with one input having two different outputs
(C) A relation where each input corresponds to exactly one output
(D) A relation with no inputs or outputs
(E) A relation with only one input
- Q2.** What is the domain of the function $f(x) = \frac{1}{x}$?
- (A) All real numbers
(B) All real numbers except $x = 0$
(C) Only positive real numbers
(D) Only negative real numbers
(E) Only $x = 1$
- Q3.** The range of the function $g(x) = x^2$ is
- (A) All real numbers
(B) All positive real numbers
(C) All negative real numbers
(D) Only $y = 0$
(E) Only $y = 1$
- Q4.** In the context of planetary motion, if d represents the distance of a planet from the sun, and t represents time, which of the following could be a function?
- (A) $d = t^2$
(B) $t = d^2$
(C) $d = 2t$
(D) $t = 2d$
(E) $d = t + 2$
- Q5.** The notation $f(2)$ represents the value of the function f at
- (A) $x = 1$
(B) $x = 2$
(C) $y = 1$
(D) $y = 2$
(E) $x = 0$
- Q6.** Which of the following statements about the function $h(x) = x^3$ is true?
- (A) The domain is all real numbers, and the range is all positive real numbers
(B) The domain is all positive real numbers, and the range is all real numbers
(C) The domain and range are both all real numbers
(D) The domain is all negative real numbers, and the range is all positive real numbers
(E) The domain and range are both limited to $x = 0$
- Q7.** For the relation given by the set of ordered pairs $\{(1, 2), (2, 3), (3, 2), (4, 5)\}$, which of the following is true?
- (A) It is a function because each input corresponds to exactly one output
(B) It is a function because it has four distinct outputs
(C) It is not a function because one input corresponds to two different outputs
(D) It is not a function because two inputs correspond to the same output
(E) It is neither a function nor a relation
- Q8.** Given the graph of a function, how can you determine if it is a function using the vertical line test?
- (A) By drawing a horizontal line that intersects the graph at more than one point
(B) By drawing a vertical line that intersects the graph at exactly one point
(C) By drawing a vertical line that intersects the graph at more than one point
(D) By drawing a diagonal line that intersects the graph
(E) By counting the number of x -intercepts

Open Ended Questions (FRQ)

- Q9.** A space mission involves launching a satellite into orbit around the Earth. The relationship between the distance d (in miles) of the satellite from the center of the Earth and the time t (in hours) after launch can be modeled by the function $d(t) = 2000 + 500t$. Determine the domain and range of this function, assuming the satellite travels in a straight line and does not change direction. Provide your answer in the space below, clearly labeling the domain and range.
- Q10.** A lab experiment measures the velocity v (in meters per second) of a particle as a function of time t (in seconds) using the equation $v(t) = 2t^2 - 5t + 1$. Evaluate $v(3)$ and explain what this value represents in the context of the experiment. Provide your calculations and explanation in the space below.

Chef's Tips

- Emphasize the importance of understanding function notation.
- Use visual aids to illustrate key concepts like the vertical line test.
- Encourage students to use real-world examples to reinforce their understanding of functions.